

LECTURE 4: MONOPOLIES AND DEMAND ELASTICITY

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WEEK 2

MONOPOLIES: LET'S BRING THE CUSTOMER INTO IT

- We've been thinking a lot about welfare implications of monopolies and drilling into how monopoly firms behave
- But there's the other side of the equation:

How do customer behaviours affect monopoly power

MONOPOLIES: LET'S BRING THE CUSTOMER INTO IT

- Imagine you're a pharmaceutical company that makes a life saving drug that only you make
- You have 100 customers that will pay anything to get this drug to keep on living
- No matter how high you raise your prices, they will keep demanding
- How high should the monopolist raise their prices?

MONOPOLIES: LET'S BRING THE CUSTOMER INTO IT

Martin Shkreli

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From Wikipedia, the free encyclopedia

Martin Shkreli (/ˈʃkrɛli/; born March 17, 1983) is an American investor and businessman. Shkreli is the co-founder of the hedge funds Elea Capital, MSMB Capital Management, and MSMB Healthcare, the co-founder and former [CEO](#) of [pharmaceutical](#) firms Retrophin and [Turing Pharmaceuticals](#), and the former CEO of start-up software company Gödel Systems, which he founded in August 2016.

In September 2015, Shkreli was widely criticized when Turing obtained the manufacturing license for the [antiparasitic](#) drug [Daraprim](#) and raised its price to insurance companies from \$13.50 to \$750.00 ([USD](#)) per pill.

In 2017, Shkreli was convicted in federal court on two counts of [securities fraud](#) and one count of [conspiracy](#). He was sentenced to seven years in prison and up to \$7.4 million in fines. In the civil [antitrust](#) case, Shkreli was fined a further \$64.6 million to be repaid to victims. In May 2022, he was released early from the [low-security federal prison](#) in [Allenwood, Pennsylvania](#). He is permanently banned from serving as an officer of any publicly traded company.

Martin Shkreli



Shkreli in 2016

Born

March 17, 1983 (age 42)^[2]

MONOPOLIES: LET'S BRING THE CUSTOMER INTO IT

We know our monopolists trade-off - but if customers don't respond to prices, it's not a trade-off!

The trade-off for raising prices is that:

- ~~Means it will sell fewer life saving drugs losing sales but decreasing costs~~
- But it will make more profit on the life-saving drugs that it does sell

The monopolist should raise prices to infinity if the customer will always buy!

WHAT IF CUSTOMERS ARE SUPER PRICE SENSITIVE?

- You own a movie theatre and have a monopoly over food and drink sold there
- But your customers are wily, the moment your prices go above marginal costs, they will revolt and smuggle in their own snacks and not buy any of yours
- Can you exert your market power?

WHAT IF CUSTOMERS ARE SUPER PRICE SENSITIVE?

If our customers are super sensitive to price - we'll lose money from trying to raise prices

The trade-off for raising prices is that:

- Means it will sell no more snacks - they will be smuggled in
- ~~But it will make more profit on the snacks that it does sell~~

Your power as a monopolist is useless if customers will no longer buy for a slightly higher price!

LET'S FORMALIZE THIS

Recall from 100A, or learn very quickly, that the price elasticity of demand is the percentage drop in demand from a 1% increase in price:

$$\varepsilon_D = \frac{\% \text{ Change in quantity}}{\% \text{ Change in Price}} = \frac{\partial Q}{\partial P} \frac{P}{Q}$$

The elasticity is negative, because an **increase** in price leads to a **decrease** in quantity demanded

LET'S FORMALIZE THIS

The more negative the elasticity, the more sensitive to price the customer is:

- $\varepsilon = 0$ The customer is perfectly inelastic, no change in demand from a change in price (our life-saving drug example)
- $-1 < \varepsilon < 0$ The customer is inelastic, a change in price leads to a relatively smaller change in demand
- $-\infty < \varepsilon < -1$ The customer is elastic, a change in price leads to a relatively large change in demand
- $\varepsilon = -\infty$ The customer is perfectly elastic, even a small increase in price completely removes all demand

The monopolist's ability to raise prices above marginal costs depends on the customer's price elasticity

FINDING THE ELASTICITY FOR LINEAR DEMAND

When demand is **linear**, the elasticity will change with quantity/price - customers are more inelastic at lower prices

Example:

$$Q = 100 - P$$

$$\varepsilon_D = \frac{dQ}{dP} \frac{P}{Q} = -1 \times \frac{P}{100 - P}$$

- When $P = 10$, $\varepsilon_D = -1 \times \frac{10}{90} = -0.11$
- When $P = 50$, $\varepsilon_D = -1 \times \frac{50}{50} = -1$
- When $P = 90$, $\varepsilon_D = -1 \times \frac{90}{10} = -9$

Note: This should make sense, a 1% change in price at \$90 is \$0.9, which should move demand a lot; but a 1% change at \$10 is only \$0.1 and should not affect demand that much

THE MONOPOLIST WILL ALSO PRICE UNTIL THE ELASTIC REGION OF DEMAND

In the inelastic region of demand (low prices)

- Raising prices leads to a proportionally smaller decrease in quantity demanded
- The profits gained from higher margins $>$ profits lost from fewer sales
- **Result:** Total profits increase $\rightarrow$ Keep raising prices

In the elastic region of demand (higher prices)

- Raising prices leads to a proportionally larger decrease in quantity demanded
- The profits gained from higher margins $<$ profits lost from fewer sales
- **Result:** Total profits start to decrease $\rightarrow$ Stop raising prices

Exactly where in the elastic region the monopoly stops will depend on the interactions of demand and cost

EXAMPLE:

Say the monopolist has zero costs and demand is:

$$Q = 100 - P$$

In the low price region:

- Price goes from \$10 to \$11
- Quantity goes from 90 to 89
- Monopolist loses \$10 of revenue from losing customers, but makes \$89 of revenue on the customers that stay
- The monopolist should definitely make this price increase!

EXAMPLE:

Say the monopolist has zero costs and demand is:

$$Q = 100 - P$$

In the high price price region:

- Price goes from \$90 to \$91
- Quantity goes from 10 to 9
- Monopolist loses \$90 of revenue from losing customers, but makes only \$9 of revenue on the customers that stay
- The monopolist is worse off from making that price increase

PRICE MARKUPS DEPEND ON ELASTICITY

We saw in our early examples that the less sensitive to price customers are, the more the monopolist could mark up prices

In general, there's a formula linking elasticity to price compared to marginal cost:

$$P = MC \times \frac{1}{1 + \frac{1}{\varepsilon}}$$

PRICE MARKUPS DEPEND ON ELASTICITY

Let's do some examples

Let a monopolist have a constant marginal cost of \$10

If we know the elasticity of demand is -1.22, let's find price

$$P = MC \times \frac{1}{1 + \frac{1}{\varepsilon}}$$

LET'S SEE THIS GEOMETRICALLY

The following are two different customer groups with different elasticities demands

But they will demand the same quantity from the monopolist



